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Individual Undergraduate Project (FYP)

Project Interim Report

Skin Care Recommendation System for Nepal

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1.0 Introduction and Context

The availability of appropriate information on skin care is becoming more and more recognized as a part of personal health care. In many countries consumer beauty markets have been booming, with the growth of personalisation platforms based on AI that offer a more personalised range of products to match each customer's skin type, receiving section of the country and the hilly regions. But Nepal is a beautiful country, and there are some significant differences between the Terai and the hilly regions.

Although there are a variety of factors that can impact the health of the skin, the majority of tools available worldwide for skin care recommendations are neither trained on data from Nepalese populations nor limited to products on the Nepalese market.

A promising way to bridge this gap is to combine the research work from computer vision and recommender system. Convolutional neural networks (CNNs) have been successfully applied in dermatological image classification and hybrid recommendation architectures with content-based and rule-based filtering in low-data environments have proved successful ([Aggarwal, 2016](#)). Combining these two paradigms into a single pipeline that uses a user selfie as input and delivers product recommendations that are available locally as output.

This report shares the interim results of developing such a system, Smart Skin Care Recommendation System for Nepal. It describes the context of the problem, existing knowledge, the purpose and objectives of the project, the selected design and methodology, as well as a feasibility and risk analysis.

1.1 Problem Statement

In Nepal, there is no reliable and accessible AI-based skin typing or skin analysis system that can suggest appropriate skin care products that can be purchased online from the local e-commerce platforms; and there is no affordable dermatology consultation available in many areas leaving individuals to rely on generic global systems and suggestions for skin care products and their use. ([Ramrakhiani and Kalbande, 2024](#); [Choudhary, 2024](#)).

Due to Nepal's varied geographical location, there is considerable difference between the climate, humidity and exposure to ultraviolet radiation which directly impacts skin health and skincare requirements. ([Goldust and Grant-Kels, 2024](#)). The climate conditions that a Kathmandu Valley resident is exposed to are different than that of someone residing in Mustang or in the high-altitude districts in the north where due to less filtering of the atmosphere, the climate is harsh with regular exposure to dry cold winds and very high exposure levels of UV radiation. On the other hand, in the lowlands of the Terai communities have to deal with the year-round presence of high ambient humidity and high temperatures. They directly affect skin condition, but none of the existing recommendation platforms considers altitude, regional air quality index or seasonal climate as factors in its suggestion logic.

The second broad dimension of the problem is one of economy of access. A large number of the internationally recommended skin care brands are not available in the Nepalese market or are extremely costly and a large mass of people cannot afford them. If a tool suggests a product that you can't buy in the country, it doesn't help you. This isn't a rare scenario, it is a common one for Nepalese skincare users interacting with AI created by foreign companies.

Lastly, dermatological access itself is limited, further accentuating the necessity of a secure on-line option. There is a high concentration of qualified dermatologists in urban area and limited or lack of qualified dermatologists in rural area. In rural areas of the hill districts and remote communities, it is difficult for people to get professional skin care advice due to its cost, time and/or practicality ([Neupane, 2012](#); [Paudel, 2020](#)). If built with the right data, the right product inventory and the right model assumptions to the population served, a self-service, AI-powered product that only needs a smartphone camera and internet connectivity is a step in the right direction to close this access gap.

2.0 Review of Existing Knowledge

The design of an Artificial Intelligence based skincare recommendation system is based on two main research areas: dermatological image classification, and recommender systems. In recent years, neural network technology, especially convolutional neural networks (CNNs), has made leaps and bounds progress in the area of skin condition detection. In practice, [\(Esteva et al. \(2017\)\)](#) showed that automated diagnosis of skin diseases using CNN-based models is comparable to dermatologists' diagnosis. Likewise, CNN architectures are also efficient in categorising various skin diseases, further clarifying their application in skin care. For more advanced architectures like EfficientNet, optimised scaling further enhances performance ([Tan and Le, 2019](#)).

However, there is still a significant challenge in dataset bias, as identified by [\(Groh et al. \(2021\)\)](#). They note that many dermatological datasets are skewed towards lighter skin tones, making it less generalizing to different populations. It is especially important to consider this in the context of Nepal as there is lack of sufficient data available in global datasets to represent the diversity of skin tones in the country.

Meanwhile, the recommender systems lay the groundwork for personalised product suggestions ([Lops et al., 2011](#)). In the case of small amount of user information, content-based filtering approaches and hybrid recommender systems have been shown to be effective. These techniques are useful for making product recommendations based on product attributes like ingredients and skin type suitability. Most of the existing systems, however, do not have localization facilities and suggest products available in the world which are not useful in Nepal.

To account for this, they systematically evaluated the performance of the models using the Fitzpatrick 17k dataset, which covers all six Fitzpatrick scale categories, and observed that models trained on the commonly used dermatology datasets performed significantly worse on prevalent in South Asian, East African, and Latin American individuals [Groh et al. \(2021\)](#). These performance deficits are not simply calibration problems, but are directly related to the fact that users who are less likely to have access to other sources of clinical guidance are the ones who are most likely to be misled.

Explainable AI, which enhances users' trust in AI, is another key concern. Techniques such as GradCAM ([Selvaraju et al., 2017](#)) can generate human-readable justifications, and techniques like provide visual explanations for CNN predictions can be used to generate explanations. But, there are not many skin care systems that include explainability features. ([Ji et al., 2023](#)).

2.1 Critical Analysis and Gap Identification

A review of published literature has shown that there are three main gaps in the literature with regards to this project.

- **Dataset bias:** Publicly available skin classification datasets are mostly comprised of images from lighter Fitzpatrick scale tones (I–III) which limits the ability of the model to generalize to skin tones found among South Asians. The Kaggle dataset provided in this project (SD20CO001) partly tackles this but it needs to be validated.
- **Market localisation:** No peer-reviewed system connects skin condition detection with a product database limited to the Nepalese market localisation. The tools that are currently available suggest luxury brands that are available around the world, but not available to most people in Nepal.
- **Explainability gap:** Many deployed skincare AI systems give recommendations without any explanations, which makes the trust of the user lower and it will not educate them about their skin. This is an issue addressed by an SLM-driven explanation layer.

2.2 Project Justification

The literature reviewed provided a definite and clear rationale for the present project. Together, the three gaps described create a problem space of academic novelty and practical importance.

Technically, the proposed system overcomes these drawbacks by using a transfer-learned EfficientNet-B2 image classification CNN, shown to be the most parameter-efficient architecture for this kind of image classification task, and a hybrid recommendation pipeline based on the content-based filtering methodology validated by and the ingredient-level matching approach [Tan and Le \(2019\)](#). Grad-CAM visualisation and its corresponding product rationales are built from scratch to solve the explainability gap of most deployed skincare artificial intelligence systems. Hence, each of the proposed architecture is associated to a specific identified gap and supported by a set of data from the literature survey [Lops et al. \(2011\)](#).

Access to dermatology is highly restricted in Nepal due to the geographic and financial reach of the dermatologist: there are only a few dermatology clinics in Kathmandu and in few urban areas in Nepal, and the majority of the population does not have easy access to dermatological expert advice regarding skin care. For people who do not want or are unable to physically consult with an expert, the internet-based AI system is a scalable and inexpensive way of accessing it.

The system's explicit constraint to products available from Nepalese e-commerce platforms ensures that recommendations are immediately actionable rather than aspirational, which represents the principal practical shortcoming of all current global skincare AI tools when applied to the Nepalese market.

3.0 Project Aims, Objectives, and Scope

3.1 Project Aim

This project seeks to create, develop and test an AI based skin care product recommender system that takes a picture of a person's face and gives them recommendations for skin care products currently available in Nepal, based on their skin type, that are explainable to them.

3.2 Project Objectives

- **Objective 1:** Build and preprocess a skin image dataset with labels. Combining the Kaggle Skin Disease dataset with any additional skin tone images from Nepal, face detection, face normalisation and face augmentation to build a training-ready skin image corpus of at least 3000 images with skin type classes.
- **Objective 2:** Train and validate a machine learning classifier that achieves a weighted F1 score of $\geq 80\%$, with transfer learning from ImageNet weights, fine-tuning on the prepared dataset and evaluation on a 15% holdout test set.
- **Objective 3:** Build a product database for a product scraper that will be used for Objective 3; ≥ 400 product SKUs scraped from Daraz, PrettyClick, Jeevee, Bonjour Nepal with names, prices, ingredients, skin-type tags, and URL (csv or structured sqlite schema).
- **Objective 4:** Develop and test skin-type recommendation system that is a hybrid of rules-based and cosine similarity ranking of ingredients vectors.
- **Objective 5:** Add an SLM explanation layer and develop a unified FastAPI endpoint: This endpoint takes as an input a base64-encoded image and optional user preferences, and returns the user's classification of skin tone, the top five recommended SLM products and their corresponding description in natural language.

3.3 Project Scope

In scope:

- Skin type pictures categorize into various skin types – dry, oily, normal, combination, acne-prone, and so on.
- All products reference from Daraz, PrettyClick, Jeevee and Bonjour Nepal.
- Limited only to still photographs of faces, not video or real-time video streaming.
- Web-based demo interface; with the possibility of native mobile application
- Support the website in English language and explanations (Nepali was not included in this phase)

Out of scope:

- Medical diagnosis or medical recommendations; system is a consumer wellness tool only.
- Begin by creating a chart for the skin diseases. Start with a chart for the skin diseases, except for the regular type classes. Collaborative filtering (requires user interaction history not available at launch).

4.0 Project Design

4.1 Design and Methods

This project follows an iterative software development methodology with machine learning model development integrated into a MLOps pipeline. The overall design comprises four major components, each with distinct methods:

The system architecture is divided into five major phases: dataset construction, skin classification model development, product database generation, recommendation engine implementation, and deployment integration. An iterative Agile-inspired workflow will be used so that each component can be independently tested and refined throughout development.

The first stage of the project involves constructing a training-ready facial skincare dataset. The primary dataset source will be the publicly available Kaggle skincare image dataset (SD20CO001), supplemented with a smaller number of locally collected images representing Nepalese skin tones and environmental conditions. The dataset will be categorised into six skin classes: dry, oily, normal, combination, acneprone, and sensitive. The skin classification component will be implemented using transfer learning with the convolutional neural network architecture. Pretrained ImageNet weights will be used as the base model, enabling the system to leverage previously learned visual features while reducing training time and computational cost.

The final classification layers will be adjusted on the skincare dataset utilizing the PyTorch deep learning framework. Class weighting will be implemented with cross-entropy loss to tackle class imbalance, and the Adam optimizer will be utilized for gradient optimization. The evaluation of model performance will involve accuracy, precision, recall, weighted F1-score, and confusion matrices. A database of skincare products specific to Nepal will be developed by gathering publicly accessible product details from leading Nepalese e-commerce websites like Daraz, PrettyClick, Jeevee, and Bonjour Nepal. Web scraping will be done using Python packages such as BeautifulSoup.

The scraper will extract product attributes including product name, brand, ingredients, price, target skin type, description, rating, and product URL. Extracted information will be cleaned and normalised before being stored in a structured SQLite database or CSV format. Ingredient names will also be standardised to reduce inconsistencies across different websites.

The recommendation engine will use a hybrid recommendation strategy combining rule-based filtering and content-based recommendation techniques. Initially, products incompatible with the predicted skin type will be filtered out using manually defined skincare rules derived from dermatological guidance and product metadata.

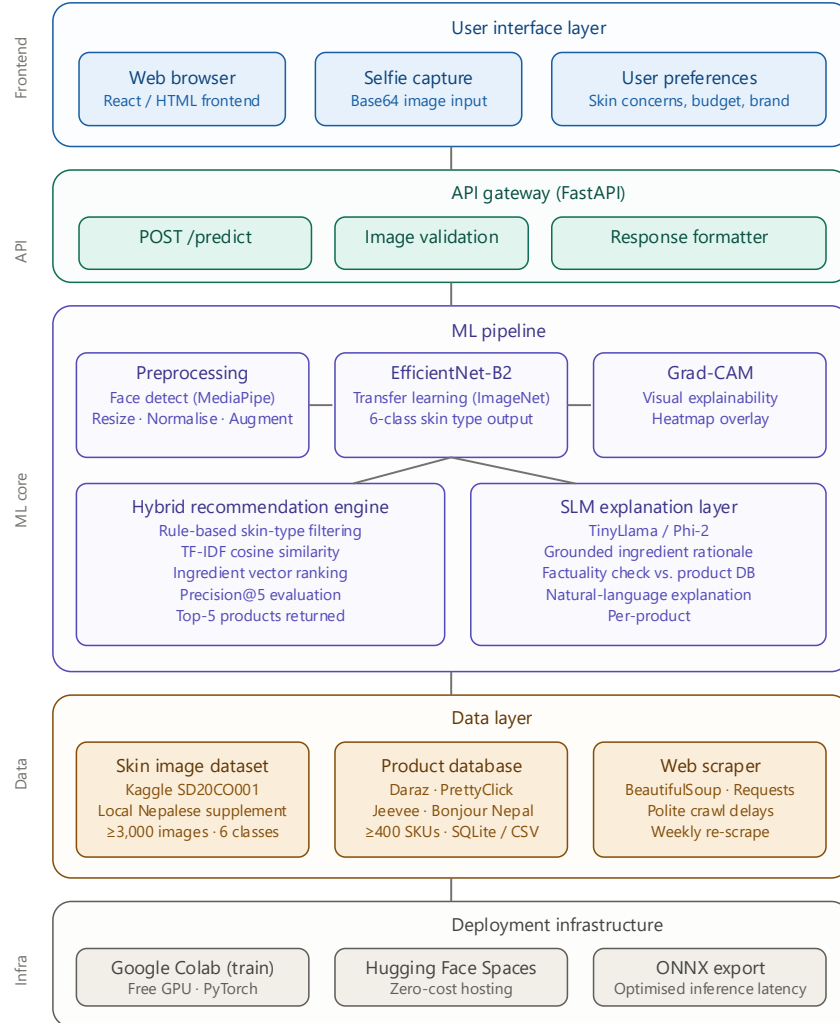


Figure 1 System Architecture Diagram

4.2 Justification of Methods

The methods chosen were selected for their optimal balance of accuracy, computational efficiency, explainability, and feasibility for an undergraduate final-year project. The training of a CNN from scratch was avoided in favor of transfer learning. The pretrained weights allows the model to learn from a smaller number of domain-specific examples, while also reducing the amount of training time and risk of overfitting.

The collaborative filtering method was ruled out for the recommendation engine and a hybrid one was chosen instead. The collaborative filtering techniques are very reliant on the larger-scale user interaction history, ratings and behavioural data which is not available during the initial deployment phase of this project. By contrast, it is possible to generate meaningful recommendations immediately with content-based filtering based on product ingredients and metadata without TF-IDF similarity. Since there are no existing publicly available datasets of skin care products specific to Nepal, web scraping is used as the main data collection technique. Manual collection is time consuming and hard to maintain, but automated scraping can achieve scalable and regularly updated product acquisition.

Finally, the deployment requirements make use of a Small Language Model (SLM) as opposed to a larger LLM. Lightweight models like TinyLlama or Phi-2 can produce concise explanations with much less memory and compute, making them more suitable to a low cost deployment environment.

4.3 Project Timeline

Project is going to be developed in iterative manner over two academic semesters. The time line includes literature review, dataset preparation, model training, recommendation system implementation, evaluation, report writing and final deployment.

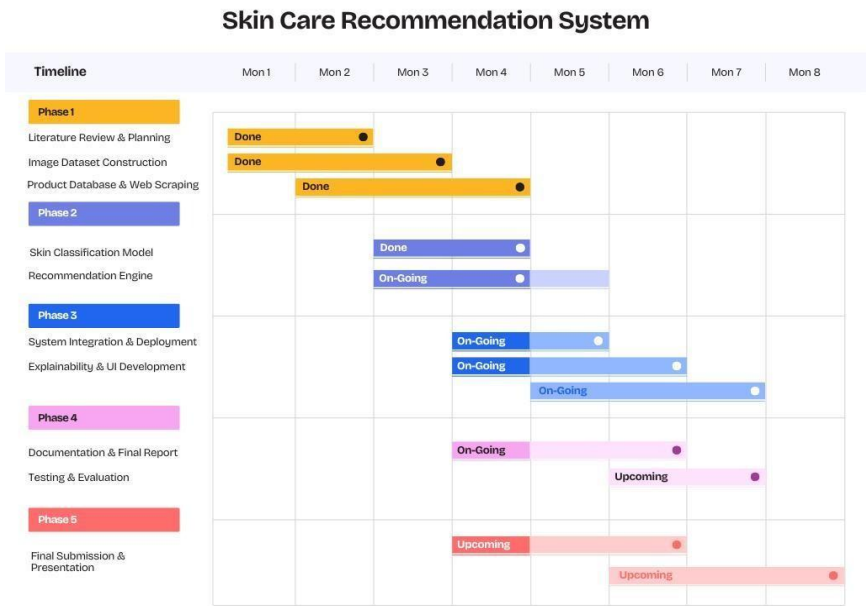


Figure 2 Gantt Chart

5.0 Feasibility, Risks, and Ethical Aspects

5.1 Feasibility

The project is considered to be feasible for the final year undergraduate project as follows. All core technologies are open-source, freely available PyTorch, Scikit-learn, FastAPI and Beautiful Soup are well documented and mature technologies.

The Kaggle dataset can be accessed by the public without special institutional approval. No server costs are required to deploy on Hugging Face Spaces. The author brings prior coursework experience to Python, machine learning and web development, minimizing learning curve with the selected technologies.

5.2 Risk Analysis and Mitigation

Potential Risk	Likelihood	Impact	Mitigation Strategy
Small or imbalanced dataset leading to overfitting	Hih	High	Apply aggressive augmentation; use classweighted loss; monitor validation /train loss gap; supplement with manually collected images.
Poor model performance on Nepalese skin tones	High	High	Collects supplementary images; report Fitzpatrick-stratified metrics; acknowledge limitation explicitly in evaluation.
SLM generates hallucinated ingredient claims	Medium	High	Ground all SLM prompts with retrieved product facts; disable free-form generation; add a factuality check step comparing output against the product record.
API inference latency exceeds acceptable response time	Low	Medium	Preprocess model; profile and optimise; Provide async processing with a result callback if desired.

Table 5.1: Risk Assessment and Mitigation

5.3 Ethical Considerations

Ethical Considerations The informed consent of all users who submit images to the system during the evaluation study will be obtained and all images collected. All images utilised will only be used for model evaluation and written consent was given and explicit to all participants. and will not be stored after the study is over. All personally identifiable data will not be kept alongside images.

The study design for the evaluation will be submitted for institutional ethics review before data collection. The project will report on performance measures broken down when ground-truth labels are available and will have a clear statement about the limitation on fairness in the final report. The system will never be able to make clinical diagnostic capability and will feature a prominent disclaimer to encourage users to seek advice from a qualified diagnostician.

It is best to see a dermatologist for medical issues. All webscraping will abide by the robots.txt rules of target platforms minimize crawl delays to 2 seconds per request, and capture only publicly available information. Data to list products is readily available. Personal data will not be retrieved from users. Scraped data will be used solely for the non-commercial academic purposes of this project. All skin type predictions will be accompanied by a level of confidence that the system is able to provide. Recommendation explanations will refer to the specific property of the ingredient and/or product that underlies each suggestion. These meaningful transparency is supported by 'measures' for the principle of AI-assisted wellness tools.

References

- Aggarwal, C.C. (2016) *Recommender Systems: The Textbook*. Cham: Springer.
- Esteva, A., Kuprel, B., Novoa, R.A., Ko, J., Swetter, S.M., Blau, H.M. and Thrun, S. (2017) 'Dermatologist-level classification of skin cancer with deep neural networks', *Nature*, 542(7639), pp. 115–118.
- Groh, M., Harris, C., Soenksen, L., Lau, F., Han, R., Kim, A., Koochek, A. and Manrai, A. (2021) 'Evaluating deep neural networks trained on clinical images in dermatology with the Fitzpatrick 17k dataset', *Proceedings of CVPR Workshops*.
- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto, A. and Fung, P. (2023) 'Survey of hallucination in natural language generation', *ACM Computing Surveys*, 55(12), pp. 1–38.
- Liu, Z., Luo, P., Wang, X. and Tang, X. (2020) 'Large-scale CelebFaces Attributes (CelebA) dataset'. Available at: <http://mmlab.ie.cuhk.edu.hk/projects/CelebA.html> (Accessed: 10 November 2024).
- Lops, P., de Gemmis, M. and Semeraro, G. (2011) 'Content-based recommender systems: State of the art and trends', in Ricci, F. et al. (eds) *Recommender Systems Handbook*. Boston: Springer, pp. 73–105.
- Neupane, S.S. (2012) 'Teledermatology in Nepal: Where are we?', *Nepal Journal of Dermatology, Venereology & Leprology*, 8(1), pp. 38–40.
- Paudel, V. (2020) 'The increasing scope of teledermatology in Nepal', *Journal of Nepal Medical Association*, 58(232), pp. 1024–1026.
- Goldust, M. and Grant-Kels, J.M. (2024) 'Using AI to help address skin health challenges caused by climate change', *International Journal of Dermatology*, 63(7), pp. e126–e127. doi: 10.1111/ijd.17222.
- Ramrakhiani, N. and Kalbande, D. (2024) 'A comprehensive review of AI-powered skincare product recommendation systems: From data collection to user experience', *Digital Health*, 10, pp. 1–18. doi: 10.1177/20427530241304073.
- Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D. (2017) 'Grad-CAM: Visual explanations from deep networks via gradient-based localization', *Proceedings of ICCV*, pp. 618–626.
- Tan, M. and Le, Q. (2019) 'EfficientNet: Rethinking model scaling for convolutional neural networks', *Proceedings of ICML*, pp. 6105–6114.
- Zhang, P., Zeng, G., Wang, T. and Lu, W. (2024) 'TinyLlama: An open-source small language model', *arXiv preprint arXiv:2401.02385*.